Valuations and p-adic distance

Instructor solutions

From factorials and prime counting to a new notion of closeness

7 problems | 100 points + 10 challenge points | Suggested time: 3–4 hours

Preparation. The lecture on Chebyshev's estimate and factorial/binomial valuations; course packet Chapters 7–8. No prior study of metric spaces is assumed.

Instructions. Write solutions on separate paper and justify your answers. You may use unique factorization, the binomial theorem, and integrality of binomial coefficients. You may use results of earlier problems even if you have not proved them. All logarithms are natural.

Notation. For a prime p and a nonzero integer m, $v_p(m)$ is the exponent of p in $|m|$. Set $v_p(0) = +\infty$, with $+\infty + t = +\infty$ for finite t. Recall that $\pi(x) = \#\{p \le x : p \text{ prime}\}$.

Grading guidance. Subpart scores total 100; the challenge contributes 10 extra points. Problems 1–3 review and apply the lecture, and Problems 4–7 introduce p-adic distance and convergence entirely on $\mathbb{Q}$. Accept equivalent proofs and carry forward arithmetic errors when the later method is correct. Do not require a construction of $\mathbb{Q}_p$ or any general theory of metric spaces.

I. Factorials, binomial coefficients, and primes

1. Count the powers of a prime

[12 points]

(a) Prove Legendre's formula by counting multiples of $p, p^2, \ldots$:

$$v_p(n!) = \sum_{j \ge 1} \left\lfloor \frac{n}{p^j} \right\rfloor.$$

Explain why multiples of p^2 must be counted more than once and why the sum has only finitely many nonzero terms. (4)

(b) Compute $v_2(50!)$, $v_3(50!)$, and $v_5(50!)$. How many trailing zeros does $50!$ have in base ten? (4)

(c) Show that $v_p((pn)!) = n + v_p(n!)$. Deduce that, for $r \ge 1$,

$$v_p((p^r)!) = 1 + p + \cdots + p^{r-1} = \frac{p^r - 1}{p - 1}.$$

(4)

Solution

Each integer $m \le n$ contributes one copy of p for each $j \ge 1$ with $p^j \mid m$. Thus summing these contributions first over m, then over j, gives the formula. A multiple of p^2 contributes at least two copies; no $p^j > n$ contributes. The requested values are

$$v_2(50!) = 25 + 12 + 6 + 3 + 1 = 47, \quad v_3(50!) = 16 + 5 + 1 = 22, \quad v_5(50!) = 10 + 2 = 12.$$

Since $10^k \mid 50!$ exactly when both 2^k and 5^k divide it, the number of trailing zeros is $\min(47, 12) = 12$. Separating the first term of Legendre's formula gives $v_p((pn)!) = n + \sum_{j \ge 1} \lfloor n/p^j \rfloor$. Apply this repeatedly, starting with $v_p(1!) = 0$, to get the geometric sum.

2. Why the middle of Pascal's triangle fits inside an lcm [16 points]

Let $n \geq 1$ and $B_n = \binom{2n}{n}$.

(a) Use Problem 1 to show that

$$v_p(B_n) = \sum_{j \geq 1} \left(\left\lfloor \frac{2n}{p^j} \right\rfloor - 2 \left\lfloor \frac{n}{p^j} \right\rfloor \right).$$

Prove that every summand is either 0 or 1. *Hint:* Write $n/p^j = q + t$, with $q \in \mathbb{Z}$ and $0 \leq t < 1$. (5)

(b) Compute $v_2\binom{20}{10}$, $v_3\binom{20}{10}$, and $v_5\binom{20}{10}$ without expanding the binomial coefficient. (3)

(c) Put $L_n = \operatorname{lcm}(1, 2, \ldots, 2n)$. Determine $v_p(L_n)$ and prove that $B_n \mid L_n$ by comparing valuations for every prime. (5)

(d) Prove that if $n < p \leq 2n$, then $v_p(B_n) = 1$. (3)

Solution

Subtract twice the valuation of $n!$ from the valuation of $(2n)!$. Writing $n/p^j = q + t$ gives $\lfloor 2(q+t) \rfloor - 2\lfloor q + t \rfloor = \lfloor 2t \rfloor \in \{0, 1\}$. For $n = 10$, Legendre's formula yields

$$v_2(B_{10}) = 18 - 2 \cdot 8 = 2, \quad v_3(B_{10}) = 8 - 2 \cdot 4 = 0, \quad v_5(B_{10}) = 4 - 2 \cdot 2 = 0.$$

The largest power of p occurring among $1, \ldots, 2n$ is p^e, where $e = \lfloor \log(2n)/\log p \rfloor$ (this is 0 when $p > 2n$). Thus $v_p(L_n) = e$. There are at most e nonzero summands in part (a), each at most 1, so $v_p(B_n) \leq e$. Unique factorization proves divisibility. Finally, $n < p \leq 2n$ gives $\lfloor 2n/p \rfloor = 1$ and $\lfloor n/p \rfloor = 0$. Also $p^2 > 2n$: since $p \geq n + 1$, $p^2 \geq (n+1)^2 > 2n$. All later terms vanish.

3. Recover the scale of prime counting [16 points]

(a) Explain why the central coefficient is largest in row $2n$, and prove $\frac{4^n}{2n+1} \leq B_n < 4^n$. *Hint:* Compare consecutive coefficients and sum the row. (4)

(b) Using Problem 2, prove

$$\log B_n \leq \pi(2n) \log(2n), \qquad \pi(2n) \geq \frac{2n \log 2 - \log(2n+1)}{\log(2n)}.$$

Explain why monotonicity of π gives a constant $c > 0$ such that $\pi(x) \geq c\, x/\log x$ for all sufficiently large real x. (6)

(c) You may take the following lecture estimate as given: $\vartheta(x) = \sum_{p \leq x} \log p \leq Ax$ for $x \geq 2$, where $A > 0$ is a fixed constant. Split the primes at $\sqrt{x}$ to prove $\pi(x) \leq \sqrt{x} + \frac{2Ax}{\log x}$ for $x \geq 4$. Deduce $\pi(x) \leq Cx/\log x$ for some fixed C and sufficiently large x. (4)

(d) Do these bounds prove $\pi(x) \sim x/\log x$? Explain the distinction between $\asymp$ and $\sim$. (2)

Solution

The ratio of consecutive coefficients is $\binom{2n}{k+1}/\binom{2n}{k} = (2n-k)/(k+1)$, so they increase up to $k = n$ and decrease afterward. Their sum is 4^n, giving the lower bound; positive terms other than B_n make the upper bound strict. For each $p \leq 2n$, Problem 2 gives $v_p(B_n) \log p \leq \lfloor \log(2n)/\log p \rfloor \log p \leq \log(2n)$. Sum over these primes and combine with part (a). For real x, take $n = \lfloor x/2 \rfloor$. Then $2n \leq x$, $2n/x \to 1$, and $\log(2n)/\log x \to 1$. The resulting lower bound, divided by $x/\log x$, tends to $\log 2$; thus $c = (\log 2)/2$ works for sufficiently large x. Every prime in $(\sqrt{x}, x]$ contributes at least $\frac{1}{2}\log x$ to $\vartheta(x)$, so $(\pi(x) - \pi(\sqrt{x}))\frac{1}{2}\log x \leq Ax$. Use $\pi(\sqrt{x}) \leq \sqrt{x}$ and then $\log x/\sqrt{x} \to 0$; for example $C = 2A + 1$ eventually works. The two bounds prove $\pi(x) \asymp x/\log x$, meaning that the ratio is bounded above and below by positive constants for large x. They do not prove $\sim$,

Solution (continued)

which requires the ratio to tend to 1.

II. Divisibility becomes a notion of distance

4. Extend the valuation to rational numbers [14 points]

For $a, b \in \mathbb{Z} \setminus \{0\}$, define $v_p(a/b) = v_p(a) - v_p(b)$; keep $v_p(0) = +\infty$.

(a) Prove that this definition is independent of the representation of a rational number as a fraction. (4)

(b) For nonzero $x, y \in \mathbb{Q}$, prove $v_p(xy) = v_p(x) + v_p(y)$ and $v_p(x^{-1}) = -v_p(x)$. Compute $v_2(12/25)$, $v_5(12/25)$, and $v_3(9/14)$. (4)

(c) Prove, for all $x, y \in \mathbb{Q}$, that

$$v_p(x + y) \geq \min\{v_p(x), v_p(y)\}.$$

Show that equality holds when $v_p(x) \neq v_p(y)$. Give nonzero x, y for which the inequality is strict and $x + y \neq 0$. *Hint:* First work with integers, factor out the smaller power of p, then use a common denominator. (6)

Solution

If $a/b = c/d$, then $ad = bc$, so $v_p(a) + v_p(d) = v_p(b) + v_p(c)$ and the two definitions agree. Multiplication and inversion follow by adding or subtracting the integer valuations of numerators and denominators. The numerical answers are $2, -2, 2$, respectively. For nonzero integers a, b, let $r = \min(v_p(a), v_p(b))$ and write $a = p^r u$, $b = p^r v$. Then $p^r \mid a + b$. If the valuations differ, exactly one of u, v is divisible by p, so $u + v$ is not divisible by p, proving equality. For rational $x = a/d$, $y = b/d$, with a common nonzero integer denominator, subtract $v_p(d)$ from the integer inequality; the equality statement also follows. Zero arguments are immediate under our convention. For strict inequality, take $x = 1$, $y = p - 1$: both have valuation 0, but $x + y = p$ has valuation 1.

5. The ultrametric inequality [14 points]

Define the *p-adic absolute value* by $|0|_p = 0$ and $|x|_p = p^{-v_p(x)}$ for $x \neq 0$. Define the *p-adic distance* by $d_p(x, y) = |x - y|_p$.

(a) Prove $|xy|_p = |x|_p |y|_p$ and the *ultrametric inequality*

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

When $|x|_p \neq |y|_p$, prove that equality holds. (6)

(b) Verify the defining properties of a metric: $d_p(x, y) \geq 0$, equality to 0 exactly when $x = y$, symmetry, and the triangle inequality. In fact, prove the stronger statement $d_p(x, z) \leq \max\{d_p(x, y), d_p(y, z)\}$. (5)

(c) Compute $|81|_3$, $|1/27|_3$, and $|10|_3$. Which is closer to 0 in d_3: 81 or $1/27$? Compare with ordinary absolute value. (3)

Solution

For nonzero factors, exponentiate the product identity from Problem 4; zero factors also satisfy it. Since $t \mapsto p^{-t}$ is decreasing, the valuation inequality becomes the ultrametric inequality, and unequal valuations give equality. The case $x + y = 0$ follows from $|0|_p = 0$. Nonnegativity and definiteness follow from the definition. Symmetry follows from $v_p(-u) = v_p(u)$. Apply the ultrametric inequality to $x - z = (x - y) + (y - z)$. Because $\max(a, b) \leq a + b$ for nonnegative a, b, the ordinary triangle inequality follows too. The three absolute values are $3^{-4} = 1/81$, $3^3 = 27$, and 1. Thus 81 is closer to 0 in d_3, whereas $1/27$ is closer in ordinary distance.

6. Closeness means agreement modulo a prime power [12 points]

(a) For integers a, b and $k \geq 0$, prove

$$d_p(a,b) \leq p^{-k} \quad \Longleftrightarrow \quad a \equiv b \pmod{p^k}.$$

(4)

(b) Compute $d_5(7,32)$, $d_5(7,132)$, and $d_5(7,8)$. Which of $32, 132, 8$ is closest to 7 in this metric? (3)

(c) List all integers b with $0 \leq b < 125$ and $d_5(7,b) \leq 5^{-2}$. Explain why the full set of integers satisfying this distance bound forms a congruence class. (3)

(d) Show that $|m|_p \leq 1$ for every integer m. Must this hold for every rational number? Give a reason or counterexample. (2)

Solution

For $a \neq b$, the distance bound is equivalent to $v_p(a-b) \geq k$, which is exactly $p^k \mid a-b$. When $a = b$, both assertions hold. The distances are $1/25$, $1/125$, and 1, respectively, so 132 is closest. The integers in part (c) are $7, 32, 57, 82, 107$: all are 7 modulo 25. The criterion asks for divisibility of a difference, which allows arbitrarily large ordinary differences. Nonzero integers have nonnegative valuation, so their absolute values are at most 1; zero also satisfies the bound. For a rational counterexample, $|1/p|_p = p > 1$.

III. First examples of p-adic convergence

Definition. A sequence (x_n) of rational numbers *converges p-adically* to $L \in \mathbb{Q}$ if, for every real $\varepsilon > 0$, there is an integer N such that $|x_n - L|_p < \varepsilon$ whenever $n \geq N$. Equivalently, the ordinary real sequence $|x_n - L|_p$ tends to 0. For integer x_n, L, Problem 6 says this happens exactly when, for every $k \geq 1$, eventually $x_n \equiv L \pmod{p^k}$.

7. Large integers can converge to zero [16 points]

(a) Prove $p^n \to 0$ p-adically. Use Problem 1 to prove $n! \to 0$ p-adically for every fixed prime p. In each case give a bound on $|x_n|_p$ that tends to 0. (5)

(b) Let $S_n = 1 + p + p^2 + \cdots + p^n$. Use the finite geometric-sum identity to prove

$$S_n \longrightarrow \frac{1}{1-p} \quad p\text{-adically.}$$

Compute the exact distance to the proposed limit. Specialize to $p = 2$: what is the limit of $1 + 2 + 4 + \cdots + 2^n$? (5)

(c) Let $T_n = \sum_{j=1}^{n} j!$. For any $m > n$, use the ultrametric inequality repeatedly to show

$$|T_m - T_n|_p \leq p^{-\lfloor (n+1)/p \rfloor}.$$

Deduce that (T_n) is *Cauchy*: for every $\varepsilon > 0$, there is N such that $|T_m - T_n|_p < \varepsilon$ for all $m, n \geq N$. You are not asked to find a rational limit. (4)

(d) Prove that a sequence of rational numbers has at most one rational p-adic limit. *Hint:* Apply the ultrametric inequality to $L - M = (L - x_n) + (x_n - M)$. (2)

Solution

We have $|p^n|_p = p^{-n} \to 0$. Legendre's formula gives $v_p(n!) \geq \lfloor n/p \rfloor$, so $|n!|_p \leq p^{-\lfloor n/p \rfloor} \to 0$. The geometric-sum identity gives

$$S_n - \frac{1}{1-p} = -\frac{p^{n+1}}{1-p}.$$

Since $p \nmid (1-p)$, the distance is exactly $p^{-(n+1)}$, which tends to 0. For $p = 2$ the limit is -1, with $S_n = 2^{n+1} - 1 \equiv -1 \pmod{2^k}$ as soon as $n + 1 \geq k$. For $m > n$, the finite-sum ultrametric inequality gives

$$|T_m - T_n|_p \leq \max_{n+1 \leq j \leq m} |j!|_p \leq p^{-\lfloor (n+1)/p \rfloor}.$$

Choose N making the last expression less than ε for $n \geq N$. Symmetry covers $m < n$, and $m = n$ gives distance 0. This proves the Cauchy assertion; it does not by itself establish a limit in $\mathbb{Q}$. The completion of $\mathbb{Q}$ for this metric is denoted $\mathbb{Q}_p$; its construction lies beyond this assignment. If L and M were both limits, then $|L - M|_p \leq \max(|L - x_n|_p, |x_n - M|_p) \to 0$. Definiteness forces $L = M$.

Optional challenge: the geometry of an ultrametric [10 extra points]

(a) For three distinct rational numbers, prove that the largest of their three pairwise p-adic distances occurs at least twice. Thus every triangle is isosceles, with its two largest sides equal (equilateral triangles are allowed). (4)

(b) For a real $r > 0$, define the closed ball $B(a, r) = \{x \in \mathbb{Q} : d_p(x, a) \leq r\}$. Prove that if $b \in B(a, r)$, then $B(b, r) = B(a, r)$. (3)

(c) Prove that two closed balls are either disjoint or one is contained in the other. *Hint:* If the balls intersect, choose a point in their intersection and apply part (b). (3)

Solution

If one distance were strictly larger than the other two, the ultrametric inequality applied to that pair through the third point would contradict it. If $b \in B(a, r)$ and $x \in B(a, r)$, then $d_p(x, b) \leq \max(d_p(x, a), d_p(a, b)) \leq r$. This proves $B(a, r) \subseteq B(b, r)$; interchange a, b for the reverse inclusion. If $B(a, r)$ and $B(b, s)$ contain a common point c, part (b) rewrites them as $B(c, r)$ and $B(c, s)$. Assuming $r \leq s$, the first is contained in the second.